

# Changing the Representation

Exterior powers and regular representations in the master formula

Computational verification note

17 July 2026

**Outcome.** Two structurally different substitutions into the finite-group master formula were tested. For  $\Lambda^2$  of the standard three-dimensional  $S_4$  representation, explicit compound matrices agree with the pairwise-eigenvalue formula. For the regular representation of  $S_3$ , explicit  $6 \times 6$  permutation matrices agree with the element-order formula. All 38 multivariate coefficients through total degree five pass, as do direct Reynolds projector checks.

## 1 The common master formula

For any finite-dimensional representation  $W$  of a finite group  $G$ ,

$$\mathcal{S}_{G,W}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a g|_W)^{-1}. \quad (1)$$

The coefficient at  $t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}$  equals

$$\dim \left( \bigotimes_{a=1}^{\ell} \text{Sym}^{\tau_a} W \right)^G, \quad (2)$$

because the coefficient is the group average of the tensor-product character, which is the trace of the Reynolds projector. Once (1) is established, a new representation requires only a correct description of its eigenvalues. This report tests two such descriptions that use very different group data.

## 2 Exterior powers

If  $g|_V$  has eigenvalues  $\lambda_1, \dots, \lambda_n$ , then the wedge vectors  $v_{i_1} \wedge \cdots \wedge v_{i_k}$  show that  $g|_{\Lambda^k V}$  has eigenvalues

$$\lambda_I = \prod_{i \in I} \lambda_i, \quad |I| = k.$$

Therefore

$$\mathcal{S}_{G, \Lambda^k V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \prod_{|I|=k} (1 - t_a \lambda_I(g))^{-1}. \quad (3)$$

This proof needs no assumption that  $G$  is a reflection group.

### 2.1 Concrete test: $\Lambda^2$ of the $S_4$ reflection representation

Start from the permutation representation of  $S_4$  on  $\mathbb{C}^4$  and restrict it to

$$V = \{(x_1, x_2, x_3, x_4) : x_1 + x_2 + x_3 + x_4 = 0\}.$$

Numerical QR orthonormalization supplies a  $4 \times 3$  basis matrix  $B$ , so the explicit standard matrices are  $B^* P_\pi B$ . The direct exterior-square matrices are constructed without diagonalizing: in the ordered basis  $e_i \wedge e_j$ , their entries are the  $2 \times 2$  minors

$$(\Lambda^2 g)_{(k,l),(i,j)} = g_{ki}g_{lj} - g_{kj}g_{li}. \quad (4)$$

The formula route instead diagonalizes only the original  $3 \times 3$  standard matrices and forms the three pairwise eigenvalue products required by (3). Thus an error in the compound-matrix construction is not reproduced in the formula route.

### 3 The regular representation

Let  $G$  act on its basis by left multiplication. If  $g$  has order  $o(g)$ , then its permutation of the  $|G|$  basis elements has  $|G|/o(g)$  cycles, each of length  $o(g)$ . Hence

$$\det(I - tg|_{\text{Reg}}) = (1 - t^{o(g)})^{|G|/o(g)}. \quad (5)$$

If  $N_m$  is the number of elements of order  $m$ , substitution into (1) gives

$$\mathcal{S}_{G,\text{Reg}}(\mathbf{t}) = \sum_m \frac{N_m}{|G|} \prod_{a \geq 1} (1 - t_a^m)^{-|G|/m}. \quad (6)$$

For coefficient extraction, use

$$[t^d](1 - t^m)^{-c} = \begin{cases} \binom{c+d/m-1}{d/m}, & m \mid d, \\ 0, & m \nmid d. \end{cases}$$

For  $S_3$ , the order counts are

$$N_1 = 1, \quad N_2 = 3, \quad N_3 = 2. \quad (7)$$

Equations (6)–(7) therefore give an integer-arithmetic formula for every multivariate coefficient. The direct route constructs all six left-regular  $6 \times 6$  permutation matrices and never consults these order counts.

### 4 Verification protocol

For each representation and every partition  $\tau$  through total degree five:

1. Construct all representation matrices explicitly.
2. Recursively compute  $h_d$  from each matrix's eigenvalues and average  $\prod_a h_{\tau_a}$  over the group.
3. Compare with the representation-specific formula: pairwise spectra in (3) for  $S_4$ , or binomial coefficient extraction from (6) for  $S_3$ .
4. Embed the normalized occupancy basis of  $\text{Sym}^2(W)$  in  $W^{\otimes 2}$ , form the explicit Reynolds matrix, and compare rank with the coefficient.

## 5 Results

| Representation                       | $ G $ | $\dim W$ | degree | checks | result      |
|--------------------------------------|-------|----------|--------|--------|-------------|
| $S_4$ on $\Lambda^2(V_{\text{std}})$ | 24    | 3        | 0–5    | 19     | <b>PASS</b> |
| $S_3$ on Reg                         | 6     | 6        | 0–5    | 19     | <b>PASS</b> |

Selected matched dimensions are

| Representation                       | (1) | (2) | (3) | (2, 1) | (2, 2) | (3, 2) |
|--------------------------------------|-----|-----|-----|--------|--------|--------|
| $\Lambda^2(V_{\text{std}})$ of $S_4$ | 0   | 1   | 0   | 0      | 3      | 1      |
| Reg of $S_3$                         | 1   | 5   | 10  | 21     | 78     | 196    |

The large mixed coefficients in the regular case provide a useful stress test: agreement is not caused merely by both routes rounding small numbers to zero.

For the exterior case,  $\text{Sym}^2(W)$  has dimension six and the Reynolds projector has rank 1, trace  $0.9999999999999992$ , and idempotence error  $8.31 \times 10^{-16}$ . For the regular case,  $\text{Sym}^2(W)$  has dimension 21; the projector has rank 5, trace  $4.999999999999999$ , and idempotence error  $3.72 \times 10^{-16}$ . These ranks agree with the (2) coefficient in each row of the table.

## 6 What these tests establish

The exterior test exercises a nonlinear eigenvalue transformation and a coordinate-level compound-matrix construction. The regular test exercises a combinatorial compression from individual group elements to element-order counts. Together they show that the master formula is representation-agnostic in precisely the claimed way: only the spectra on the chosen representation change.

The tests are finite and numerical; the all-degree statements (3) and (6) follow from the preceding eigenvalue and cycle proofs. Numerical acceptance requires the direct average to be within  $3 \times 10^{-7}$  of an integer. The projector residuals are at machine precision.

## 7 Reproducibility

Run from the repository root:

```
PYTHONPATH=for_this_guy/matrix_group_formula_verification \
.venv/bin/python \
for_this_guy/matrix_group_formula_verification/\
representation_variants/verification.py
```

The marimo notebook allows the maximum degree to be changed interactively and displays every coefficient comparison and both projector diagnostics.